

Let's create a similar design for a **PATIENT** entity type that tracks a patient's history of **medical treatments**. Each entry for patient should track the hospital where the treatment was given, the start and end dates of the treatment, any procedures performed (if any) on the basis of treatment, and a list of medications prescribed (if any). Each procedure includes the procedure name and the date it was performed. Each medication entry contains the medication name, dosage, and duration.

Identify the Entity, attribute and the relation and the cardinality.

A **sales representative**, such as **Alex**, is identified by a unique **Rep_ID**, and each representative has a **Name** (consisting of *First_Name* and *Last_Name*), a **Hire_Date**, and contact information including **Phone Numbers** (which can be multiple, such as *office phone* and *mobile phone*) and an **Email address**. Each sales representative is supervised by exactly **one other representative**, identified through **Supervisor_ID**, while they themselves may **supervise several other representatives**. Alex also handles several **clients**. Each **client** has a **Client_ID**, **Client_Name**, and an **Address** (composed of *Street*, *City*, and *Zip_Code*). A client also has a **Contact_Number** and is assigned to one **sales representative** through the **Rep_ID**. Clients can place multiple **orders**. Each **order** is identified by an **Order_ID** and includes attributes such as **Order_Date** and **Total_Amount**. Every order is associated with exactly **one client** but may include multiple **stock items**. Each **stock item** has an **Item_ID**, **Item_Name**, **Description**, **Unit_Price**, and **Manufacture_Details** (which further includes *Batch_Number* and *Date_Produced*). Stock items can appear in multiple **orders**, creating a many-to-many relationship. Each **stock item** is composed of several **components**. Every **component** is identified by a **Component_ID**, has a **Component_Name**, **Cost_Per_Unit**, and **Specifications** — which can include multiple details like *Material_Type* and *Dimension* (making this a multivalued attribute). A component can be used to assemble different **stock items**. The **assembly process** involves multiple **workers** collaborating to build the stock items. Each **worker** has a unique **Worker_ID**, a **Name** (split into *First_Name* and *Last_Name*), an **Experience_Level**, and a **Skill_Set** (which is multivalued, containing skills like *Welding*, *Assembling*, *Testing*). Each worker may work on many stock items, and each stock item may involve multiple workers. Finally, **components** are supplied by various **vendors**. Each **vendor** is identified by a **Vendor_ID** and has a **Vendor_Name** and an **Address** (which includes *Street*, *City*, *State*, and *Zip*). Vendors also have multiple **Contact_Details** such as *Phone* and *Email* (a multivalued attribute). Each vendor may supply many components, and each component can be sourced from multiple vendors.